

6.4

A Eurodollar futures price changes from 96.76 to 96.82. What is the gain or loss to an investor who is long two contracts?

The Eurodollar futures contract is designed such that each basis point (0.01) move corresponds to a change in \$25 in the contract. So with a $96.82 - 96.76 = 0.06$ change for 2 contracts at \$25 per contract, this totals to

$$6 \cdot 2 \cdot \$25 = \boxed{\$300 \text{ gained}}$$

6.6

The 350-day LIBOR rate is 3% with continuous compounding and the forward rate calculated from a Eurodollar futures contract that matures in 350 days is 3.2% with continuous compounding. Estimate the 440-day zero rate.

6.9

It is May 5, 2014. The quoted price of a government bond with a 12% coupon that matures on July 27, 2024, is 110-17. What is the cash price?

We know to find the cash price, we simply need to take the sum of the quoted price and the accrued interest since the last coupon date. The previous coupon date would have been January 27. There are 98 days between January 27 and May 5, and there are 181 days between January 27 and July 27. Using this, we can calculate the accrued interest for the days so far:

$$\frac{12}{2} \cdot \frac{98}{181} = 6 \cdot 0.54143646408 = 3.24861878448$$

With the quoted price being $110 \frac{17}{32} = 110.53125$, we know the cash price to be

$$110.53125 + 3.24861878448 = 113.779868784 \approx \boxed{\$113.78}$$

6.11

It is July 30, 2015. The cheapest-to-deliver bond in a September 2015 Treasury bond futures contract is a 13% coupon bond, and delivery is expected to be made on September 30, 2015. Coupon payments on the bond are made on February 4 and August 4 each year. The term structure is flat, and the rate of interest with semiannual compounding is 12% per annum. The conversion factor for the bond is 1.5. The current quoted bond price is \$110. Calculate the quoted futures price for the contract.

6.14

Suppose that the 300-day LIBOR zero rate is 4% and Eurodollar quotes for contracts maturing in 300, 398 and 489 days are 95.83, 95.62, and 95.48. Calculate 398-day and 489-day LIBOR zero rates. Assume no difference between forward and futures rates for the purposes of your calculations.

6.21

The three-month Eurodollar futures price for a contract maturing in six years is quoted as 95.20. The standard deviation of the change in the short-term interest rate in one year is 1.1%. Estimate the forward LIBOR interest rate for the period between 6.00 and 6.25 years in the future.

